

Exp. No: 4 Analysis of Continuous Charge Distribution in electric field. To find the (i) line charge density, (ii) Surface charge density and (iii) Volume charge density for any specific (a) line, (b) surface or (c) volume.

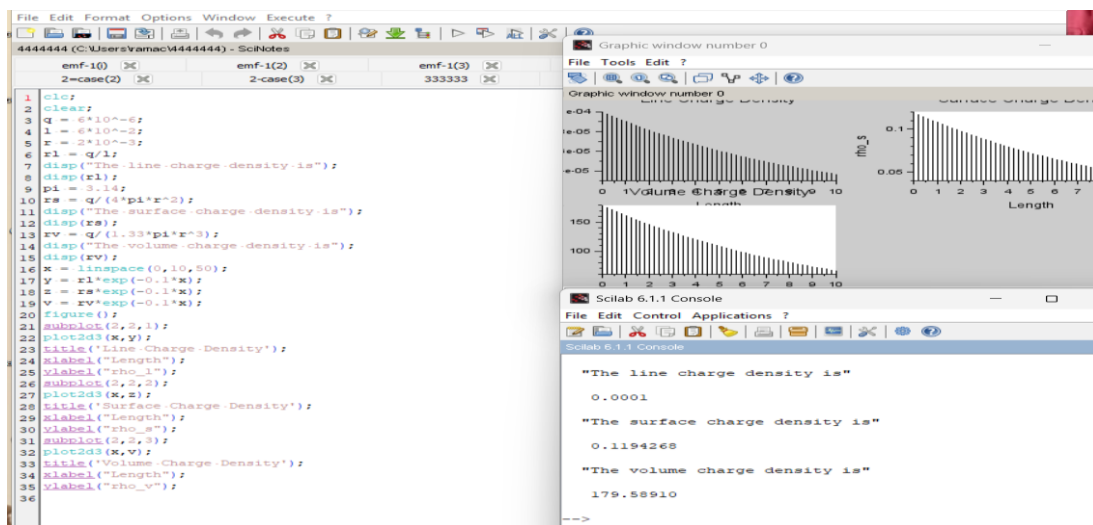
AIM:

To write program for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge.

Software required:

•SCILAB version 6.0.1

Practical Calculations:



Theoretical Calculations:

Handwritten calculations for the theoretical values of charge densities:

④

charge (q) = 6×10^{-6}

length (l) = 6 cm

radius (r) = 2 mm

$r_1 = \frac{q}{l} \Rightarrow \frac{6 \times 10^{-6}}{6 \times 10^{-2}} \Rightarrow 1 \times 10^{-4} \text{ C/m}$

$\rho_s = \frac{q}{4\pi r^2} \Rightarrow \frac{6 \times 10^{-6}}{4 \times 3.14 \times (2 \times 10^{-3})^2} \Rightarrow 0.1194 \text{ C/m}^2$

$\rho_v = \frac{q}{1.33 \times \pi \times r^3} \Rightarrow \frac{6 \times 10^{-6}}{1.33 \times 3.14 \times (2 \times 10^{-3})^3} \Rightarrow 179.69 \text{ C/m}^3$

Result: Thus the program was verified for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB was successfully.